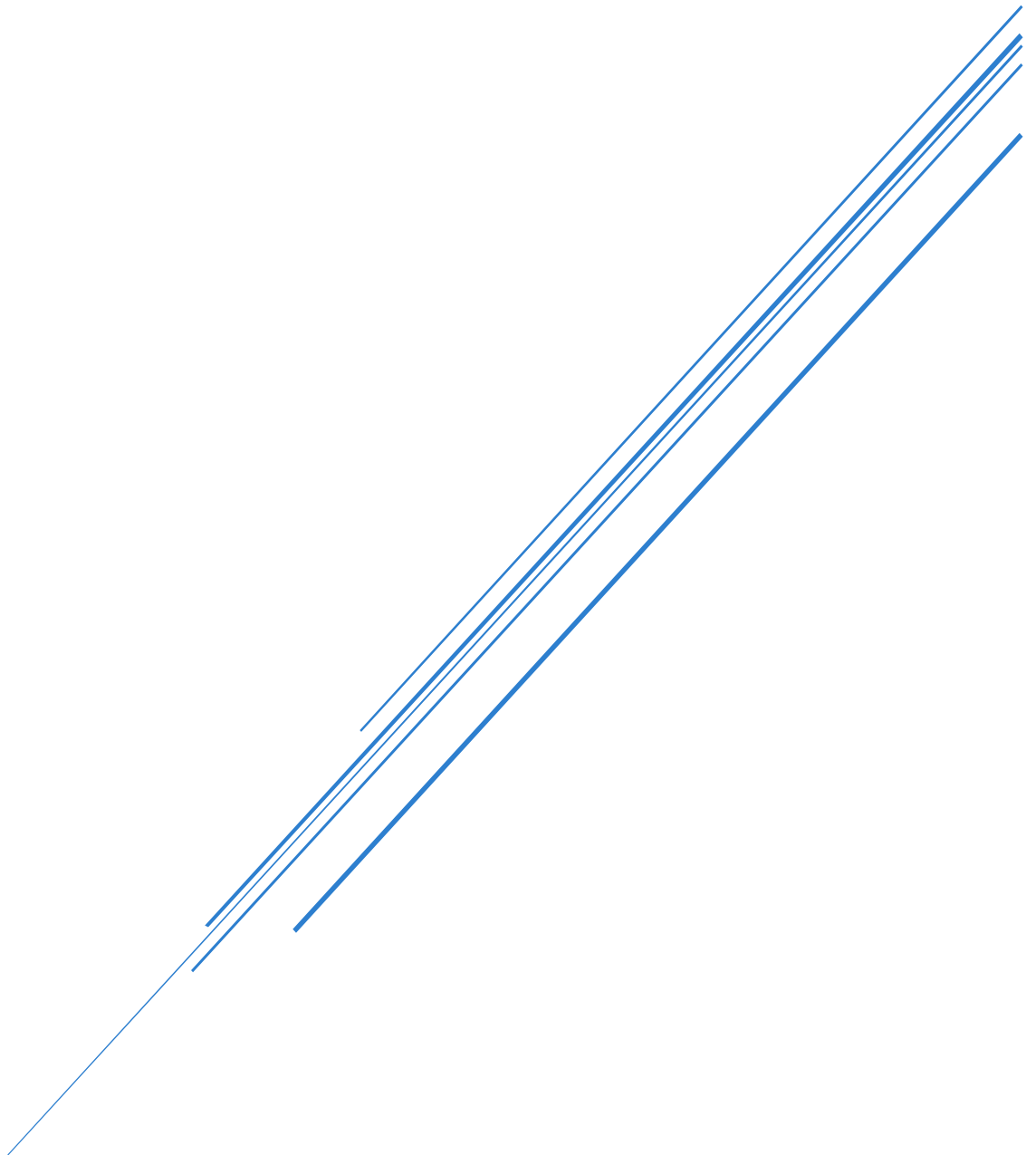


PROG7313 POE PART 3

IIE EMERIS PIETERMARITZBURG



03/06/2026

Group Members

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Welcome to the KTP Budgeting App!

What is the KTP Budgeting App?

The main purpose of the KTP Budget Tracker App is a personal finance management app which is designed to help anyone to monitor and control their everyday spending. The app allows users to record their expenses, organize them into categories, set monthly spending goals, track their financial activities over time and reward-based motivation.

Our app also uses Firebase Authentication to ensure secure login and registration, and Firebase Realtime Database to store and retrieve user data in real time. This ensures that user information is saved securely and updated instantly across the application.

Basic Features of the app:

- Log and categorize daily expenses
- Set minimum and maximum budget goals
- View past spending records
- Analyse spending trends through charts
- Receive motivational rewards for financial discipline
- Access a chatbot assistant for guidance and FAQs

Special features of the app:

Chatbot

The application includes a chatbot feature designed to assist users with common budgeting and expense-tracking queries. The chatbot operates using a set of predetermined questions and responses, allowing users to receive instant guidance and support within the application. When a user asks a question that matches one of the predefined queries, the chatbot provides a relevant answer to the question.

Badge Rewards system

The reward system was implemented to encourage users to manage their finances effectively. Users can earn achievement badges by completing financial goals and challenges within the application, such as staying within a budget, reaching savings targets, and consistently tracking expenses. A Rewards page displays each achievement, showing whether it is locked or earned, along with progress bars that

indicate how close the user is to unlocking each reward. If a user exceeds their budget or fails to meet the required conditions, the reward remains locked and is not awarded.

Design Considerations and User Interface (UI) Design

The KTP Budgeting app has 8 screens

- Login screen
- Register screen
- Home screen
- Expense entry screen
- Records screen
- Report screen
- Rewards screen
- Chatbot help screen

The UI design is very clean and very easy to read and understand. The texts, buttons and functions of each page are inside a card layout to make the app look very professional. A bottom navigation bar is used across all major screens to allow quick and consistent navigation between features. This improves usability and reduces user confusion.

GitHub

GitHub was used as the primary version control system for the development of the budgeting application. It allowed for the code to be shared amongst the team members and to be securely stored in the cloud.

GitHub Actions

GitHub Actions was used to introduce continuous integration (CI) into the project development process. This automation tool helped ensure that our application remained stable and buildable after each update was done to the project.

The App Icon:

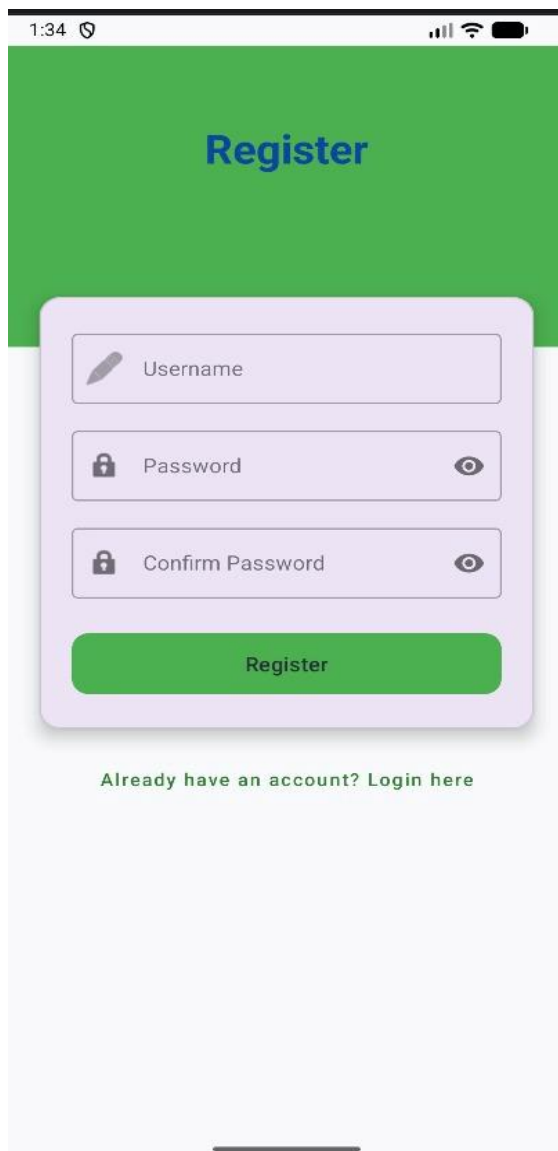


(Looka, 2026)

Login Page:

The screenshot shows a mobile app interface. At the top, there's a status bar with the time "1:33" and various icons. Below that is a green header with the text "Welcome To Budget Tracker" in white. The main content area is white and contains a login form. The form has two input fields: "Username" with the text "redman@g.com" and "Password" with the text "redman". Below the password field is a green button with the text "Click here to login in". At the bottom of the form, there's a link that says "Don't have an account? Click Here".

Register Page:



1:34

Register

Username

Password

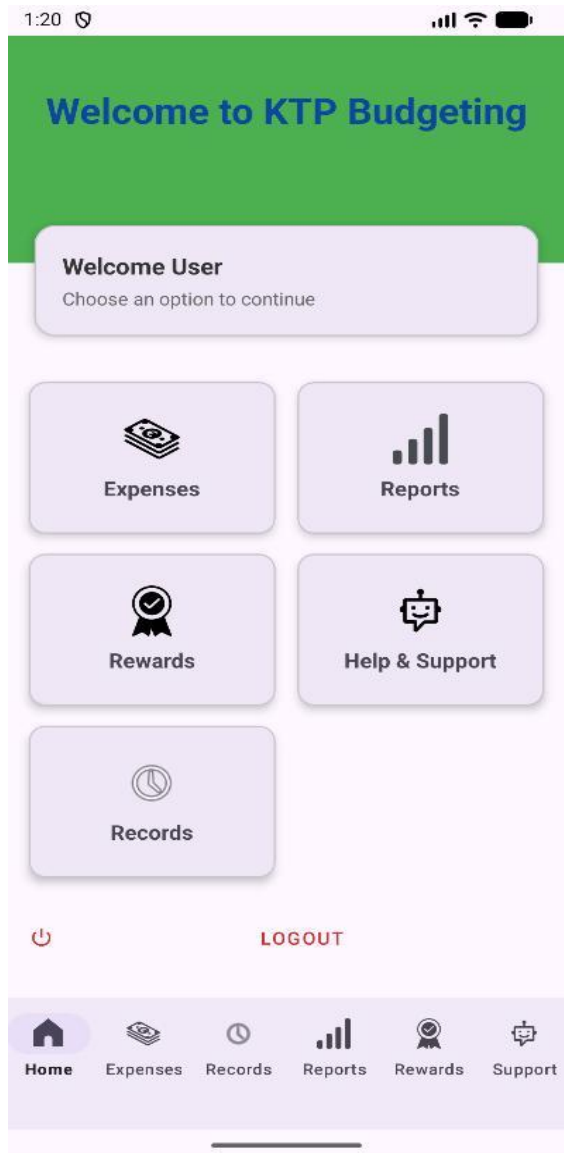
Confirm Password

Register

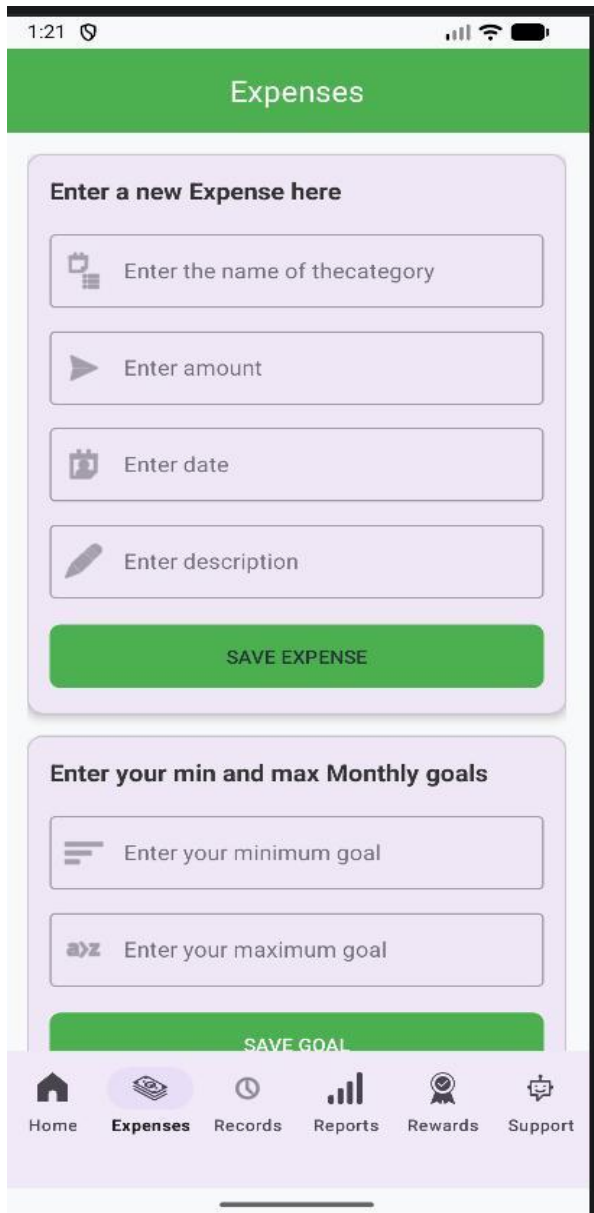
Already have an account? [Login here](#)

This mockup shows a registration form on a mobile app. The form is centered on a light green background. It has three input fields for 'Username', 'Password', and 'Confirm Password', each with a corresponding icon (pencil, lock, and eye). A green 'Register' button is at the bottom of the form. Below the form, there is a link 'Already have an account? Login here'.

Home Page:



Expenses Page:



A screenshot of the 'Expenses' page in a mobile app. The page has a green header with the title 'Expenses'. Below the header, there is a section titled 'Enter a new Expense here' with four input fields: 'Enter the name of the category' (with a tag icon), 'Enter amount' (with a right arrow icon), 'Enter date' (with a calendar icon), and 'Enter description' (with a pencil icon). A green 'SAVE EXPENSE' button is at the bottom of this section. Below this is another section titled 'Enter your min and max Monthly goals' with two input fields: 'Enter your minimum goal' (with a list icon) and 'Enter your maximum goal' (with an 'a-z' icon). A green 'SAVE GOAL' button is at the bottom of this section. The bottom navigation bar has six icons: Home, Expenses (selected), Records, Reports, Rewards, and Support.

1:21

Expenses

Enter a new Expense here

Enter the name of the category

Enter amount

Enter date

Enter description

SAVE EXPENSE

Enter your min and max Monthly goals

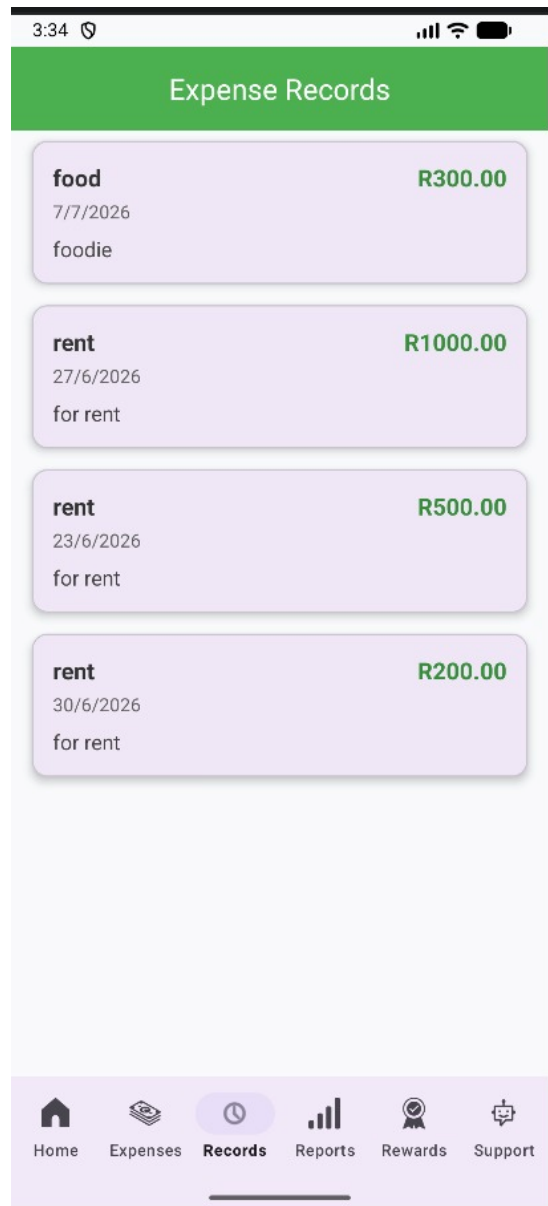
Enter your minimum goal

Enter your maximum goal

SAVE GOAL

Home Expenses Records Reports Rewards Support

Record Page:



A screenshot of the 'Expense Records' page in a mobile app. The page has a green header with the title 'Expense Records'. Below the header, there are four expense record cards. Each card displays the category, date, description, and amount. The bottom navigation bar has six icons: Home, Expenses, Records (selected), Reports, Rewards, and Support.

3:34

Expense Records

food R300.00
7/7/2026
foodie

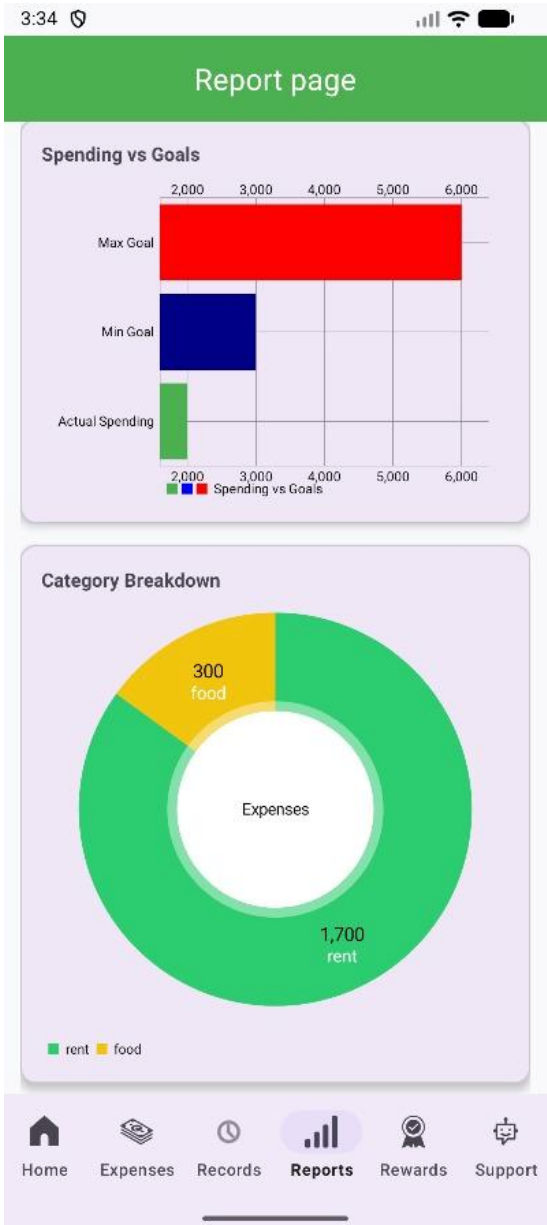
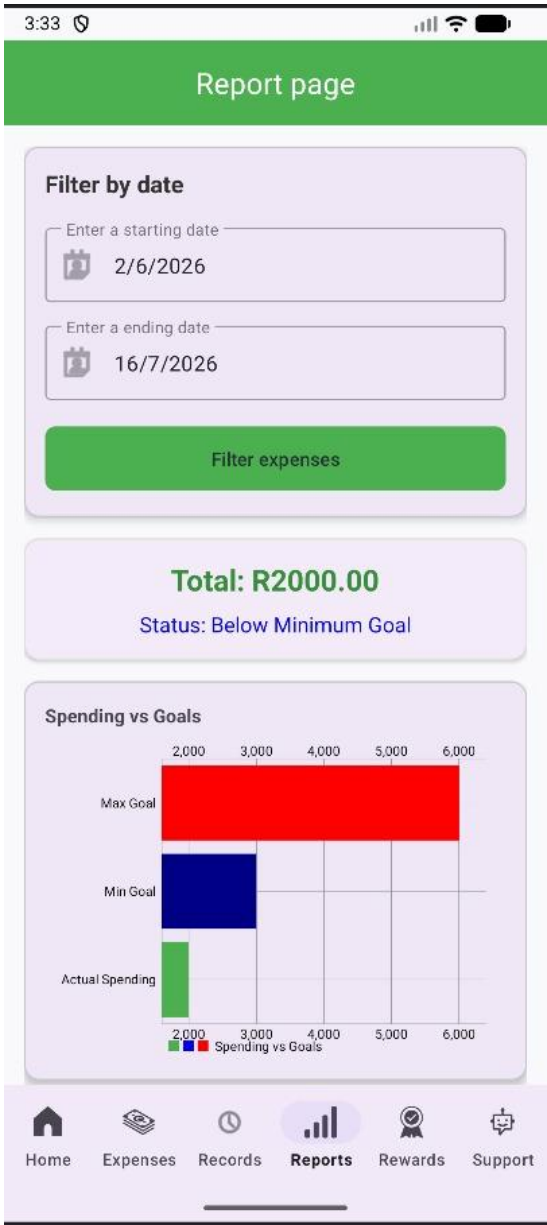
rent R1000.00
27/6/2026
for rent

rent R500.00
23/6/2026
for rent

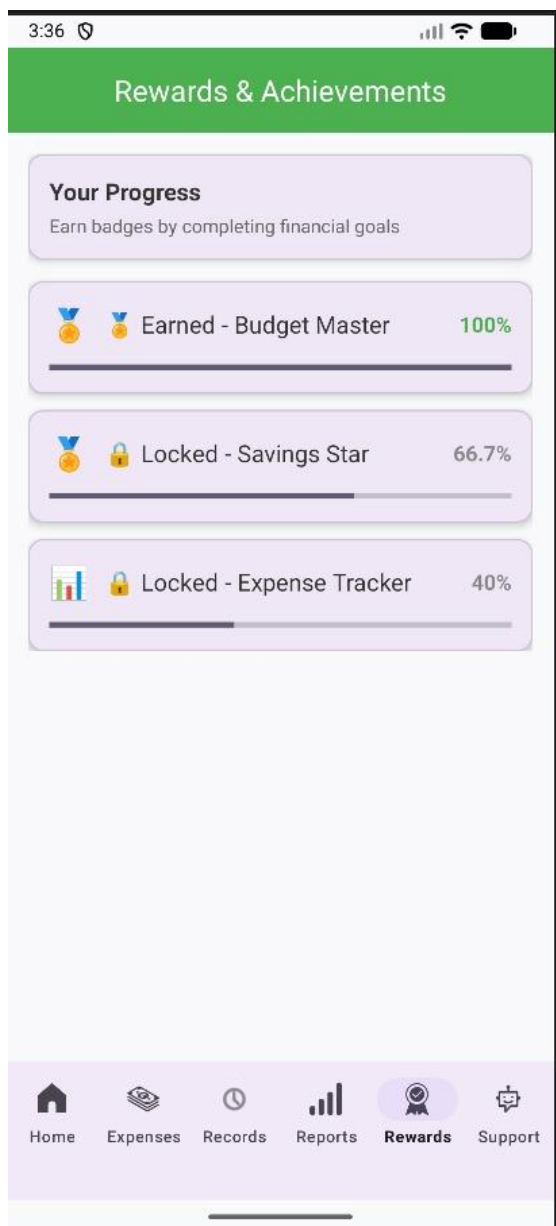
rent R200.00
30/6/2026
for rent

Home Expenses Records Reports Rewards Support

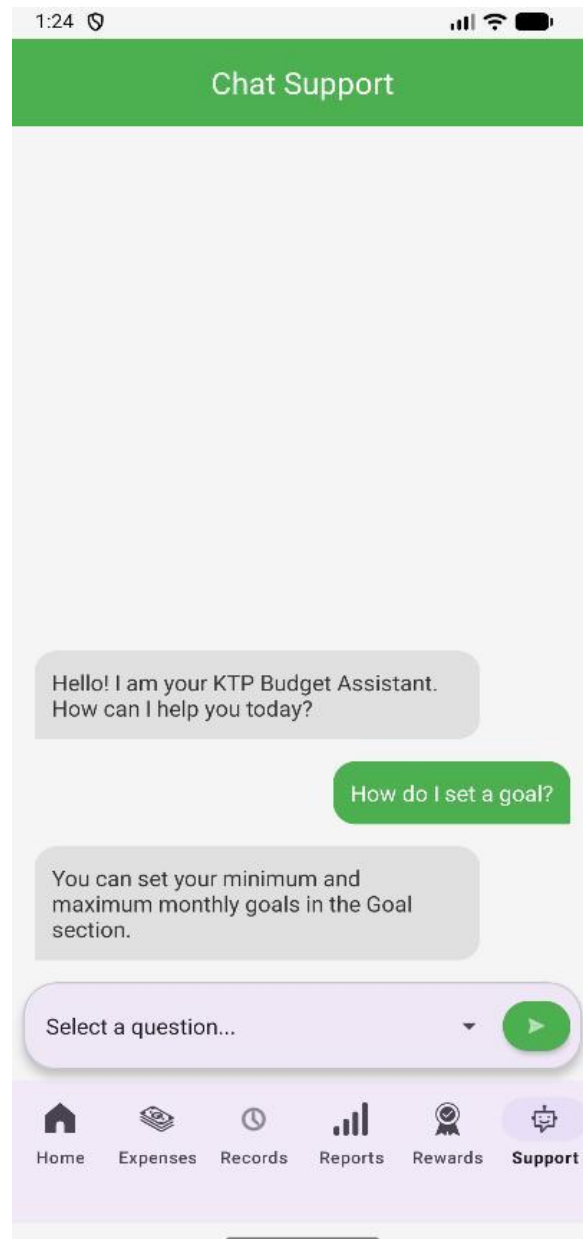
Report Page:



Rewards Page:



Chatbot page:



Database Firebase Authentication:

 **Firebase**

 Search for products

 Project Overview

 Settings >

What's new

 Phone Verification NEW

Project shortcuts

 **Authentication**

 Realtime Database

 Storage

Product categories

Databases & Storage >

Security >

AI services >

Hosting & Serverless >

DevOps & Engagement >

Spark
No-cost (\$0/month)

Upgrade
NEW

PROG7313 KTP Budgeting App ▾

Authentication

[Users](#) | [Sign-in method](#) | [Templates](#) | [Usage](#) | [Settings](#) | [Extensions](#)

 The following Authentication features will stop working when Firebase Dynamic Links shuts down soon: email link authentication for mobile apps, as well as Cordova OAuth support for web apps. ▾

 Search by email address, phone number, or user UID

Add user  

Identifier	Providers	Created ▾	Signed in	User UID
blue@red.com		Jun 3, 2026	Jun 3, 2026	CGAGHGgcJIQgj9USe3nQGB...
hello@world.com		Jun 2, 2026	Jun 2, 2026	BpD0IXWz5VYPWkOMFqh78e...
redman@g.com		Jun 1, 2026	Jun 1, 2026	58cJgHITzpN9S4ExtYeiCUsiG...
kolz@gmail.com		May 30, 2026	May 30, 2026	9ncbArfaUUhwlp2P1q3vKkau...
kiki@gmail.com		May 30, 2026	Jun 3, 2026	PmG3PLedpRNRYluONwqv6P...
test1@gmail.com		May 23, 2026	Jun 2, 2026	hf97gvCi1yehhXFhnrR3R3Buy...

Rows per page: 50 ▾ 1 - 6 of 6 < >

10 | Page

Firestore Realtime Database:

 **Firebase**

 Search for products

 Project Overview

 Settings

What's new

 Phone Verification

NEW

Project shortcuts

 **Realtime Database**

 Storage

 Authentication

Product categories

Databases & Storage

Security

AI services

Hosting & Serverless

DevOps & Engagement

Analytics

Spark

No-cost (\$0/month)

Upgrade

NEW

PROG7313 KTP Budgeting App

Realtime Database

◆ Need help with Realtime Database? Ask Gemini

Data

Rules

Backups

Usage

Extensions

Protect your Realtime Database resources from abuse, such as billing fraud or phishing

[Configure App Check](#)

×

https://prog7313-ktp-budgeting-app-default-rtdb.firebaseio.com

https://prog7313-ktp-budgeting-app-default-rtdb.firebaseio.com/

users

9ncbArfaUUhwlp2P1qi3vKkayk2

CGAGHGgcJIQgj9USe3nQG8sxNkE2

expenses

-OuCB0SM3gm5LXS40blh

amount: 2000

category: "food"

date: "3/6/2026"

description: "saving on food"

id: "-OuCB0SM3gm5LXS40blh"

goals

maxGoal: 2000

minGoal: 1200

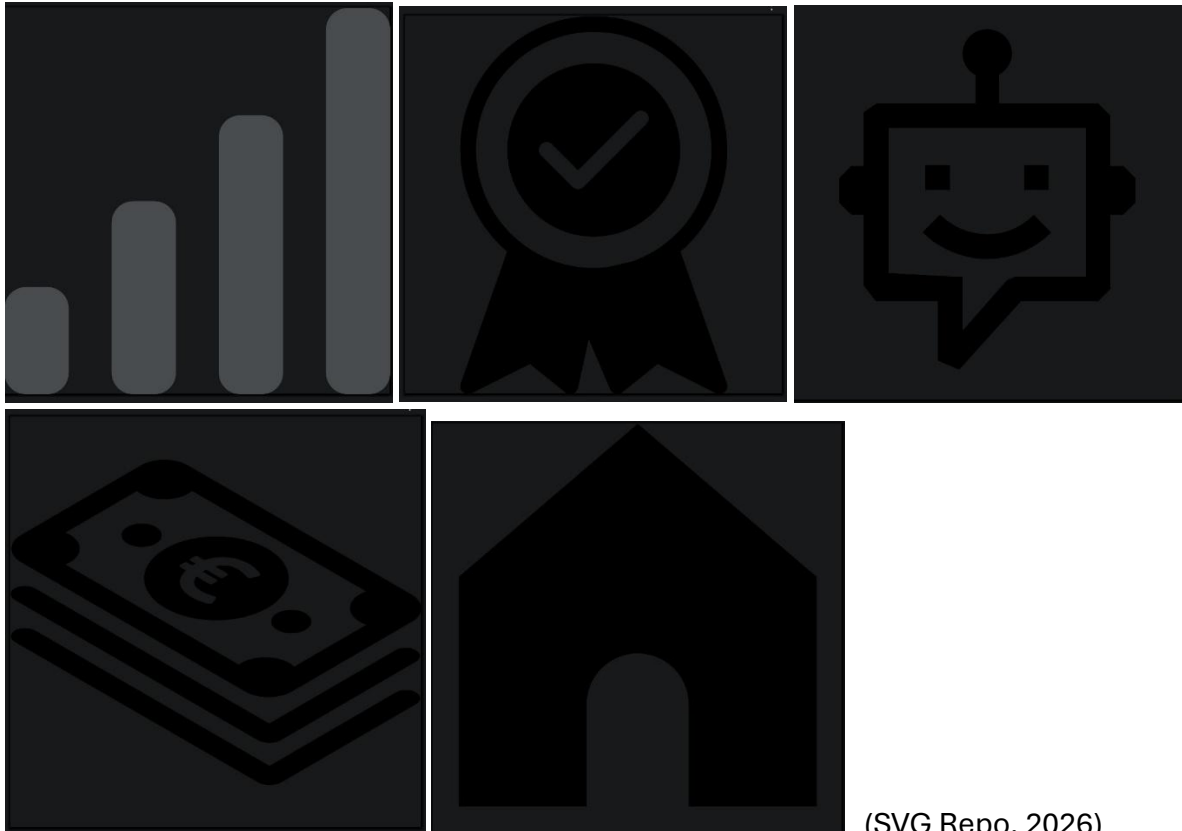
PmG3PLedpRNRyIu8Nwqv6PNbFA12

hF97gvC11yehhXFhnrR3R3BuySA3

Database location: United States (us-central1)

11 | Page

Icons used in the App:



(SVG Repo, 2026)

Reference List:

Looka, 2026. Design your own beautiful brand. [electronic print] Available at: < <https://looka.com/> > [Accessed 02 June 2026].

SVG Repo, 2026. 500.000+ Open-licensed SVG Vector and Icons. [electronic print] Available at: < <https://www.svgrepo.com/> > [Accessed 02 June 2026].